

Exact regularization thresholds for integer quadratic transport

Marcelo Gallardo^{1*}

^{1*}Department of Economics, University of California, Berkeley, Berkeley, California, USA.

Corresponding author(s). E-mail(s): marcelogallardob21@berkeley.edu;

Abstract

We study the regularization level at which a unique linear-optimal transport plan ceases to minimize a heterogeneous quadratic objective. Although integer margins make every transportation vertex integral, continuous and integer regularization generally leave that vertex at different parameter values. Using the classical optimality criterion for convex-cost flows, we express both thresholds through feasible residual cycles. The integer formula contains an explicit finite-step correction and yields a strictly larger threshold whenever the continuous threshold is finite. Under population replication, the normalized integer threshold decreases to the continuous threshold. We identify its first-order correction and prove an exact formula valid after finitely many changes of the minimizing cycle. A dependent-rounding argument also gives a sharp second-order value bound and a uniform first-order distance bound as the grid is refined. Small examples attain these rates and exhibit a change of the threshold-determining cycle. A final sensitivity result characterizes unit adjustments when one connection cost varies.

Keywords: Integer transport, Quadratic regularization, Convex-cost flows, Parametric optimization, Discrete sensitivity

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1 Introduction

Quadratic regularization makes a linear transport problem strictly convex, but its optimizer may be fractional even when the margins are integer. When transport units are indivisible, this raises a parametric question: at what regularization level does an integer allocation first change, and how does that level compare with the continuous threshold? An objective approximation alone does not determine the location of such transitions.

We answer this question for a unique linear-optimal transport plan and positive, heterogeneous quadratic weights. The main result is an exact cycle formula for its persistence threshold on each integer grid. The formula separates the continuous directional derivative from the quadratic cost of a finite step. This correction makes the discrete threshold strictly larger than its continuous counterpart. Under population replication, we identify the cycle that eventually determines the threshold and obtain an exact formula for all sufficiently large replication factors. The normalized threshold error is of first order in the grid size, whereas the optimal value error is of second order. Both rates are attained in a two-by-two example. A specialization characterizes pure group assignments under arbitrary heterogeneous weights and gives explicit sufficient conditions for their optimality.

The optimization machinery is classical. Separable convex-cost flows admit marginal-cost algorithms and residual-cycle certificates [1–3]; discrete convexity also describes sensitivity to costs and capacities [4]. Our contribution concerns the finite-step correction to a regularization threshold and its behavior under replication. We use the established flow criteria explicitly, and treat rounding and coordinate sensitivity as supporting consequences.

For continuous problems, González-Sanz and Nutz [5, Theorem 2.5] characterize the exact threshold of quadratic regularization over an arbitrary polytope. Their result allows any inner product and therefore includes heterogeneous quadratic weights. The continuous formula below is its transport-cycle specialization. The additional term in the integer formula determines a different critical cycle in some instances and yields the replication correction studied here. Feasible rounding uses the marginal and degree preservation underlying dependent rounding [6]; no new rounding algorithm is needed.

The model of Gallardo et al. [7] penalizes deviations from prescribed margins. Here the margins hold exactly, and the analysis compares continuous and indivisible flows for the same objective. Section 2 states the required flow criterion. Sections 3 and 4 give the threshold formulas and replication law. Sections 5 and 6 establish the approximation and sensitivity bounds. Appendix A records existence, the continuous solution and its derivatives, and the general two-by-two formula.

2 Transport grids and the flow criterion

Let $\mu \in \mathbb{Z}_{++}^n$ and $\nu \in \mathbb{Z}_{++}^m$, with $n, m \geq 2$ and $\sum_i \mu_i = \sum_j \nu_j$. Write

$$P = \left\{ x \in \mathbb{R}_+^{n \times m} : \sum_j x_{ij} = \mu_i, \quad \sum_i x_{ij} = \nu_j \right\}. \quad (1)$$

For $q \in \mathbb{Z}_{++}$, set $P_q = P \cap q^{-1}\mathbb{Z}^{n \times m}$. Thus P_1 consists of integer transport plans, and $qx \in qP_q$ transports integer populations $q\mu, q\nu$. The grid sets need not be nested as q increases. Index connections by $e = (i, j)$ and consider

$$F_t(x) = c \cdot x + tQ(x), \quad Q(x) = \sum_e b_e x_e^2, \quad b_e > 0, \quad t \geq 0. \quad (2)$$

The continuous optimizer $x^{\mathbb{R}}(t)$ is unique for $t > 0$. Grid optima exist because P_q is nonempty and finite; strict convexity does not imply their uniqueness.

For $x \in P_q$, a feasible signed simple cycle g alternates between origin and destination nodes. Its nonzero entries are $+1$ or -1 , and $g_e = -1$ requires $x_e > 0$. Consequently, $x + g/q \in P_q$. We exclude the two-arc walk using the same connection in both directions, which makes no change in the plan.

Lemma 1 (Classical convex-flow criterion) *For $F(x) = \sum_e (c_e x_e + a_e x_e^2)$, $a_e > 0$, a plan $x \in P_q$ is optimal if and only if*

$$F(x + g/q) - F(x) \geq 0 \quad (3)$$

for every feasible signed simple cycle g . It is the unique optimum if and only if all these inequalities are strict. Any two optima differ by at most $1/q$ in each coordinate.

Proof. Use forward residual costs $c_e + a_e(2x_e + 1/q)$ and, when $x_e > 0$, reverse costs $-c_e - a_e(2x_e - 1/q)$. Their sum along g is $q[F(x + g/q) - F(x)]$. If no cycle has negative cost, shortest-path potentials give numbers u_i, v_j such that

$$\begin{aligned} c_{ij} + a_{ij}(2x_{ij} - 1/q) &\leq u_i + v_j \leq c_{ij} + a_{ij}(2x_{ij} + 1/q) & (x_{ij} > 0), \\ u_i + v_j &\leq c_{ij} + a_{ij}/q & (x_{ij} = 0). \end{aligned}$$

These inequalities say that x_{ij} minimizes $c_{ij}z + a_{ij}z^2 - (u_i + v_j)z$ over $z \in q^{-1}\mathbb{Z}_+$, where $\mathbb{Z}_+ = \{0, 1, \dots\}$. Summation cancels the potential terms for every feasible competitor and proves optimality. Necessity follows from a negative cycle. Every other optimum must minimize the same one-dimensional quadratics, whose minimizers are at most two adjacent grid points. A zero-cost cycle gives a distinct optimum. Conversely, decomposing the difference of two optima into cycles conforming to its signs and using convexity shows that distinct optima require a zero-cost cycle. $\square$

Lemma 1 is the quadratic specialization of the standard convex-flow criterion [3, Section 12]. Increasing discrete marginal costs also give an exact minimum-cost flow formulation by expanding each connection into unit-capacity arcs. This expansion is pseudopolynomial in the total population; scaling methods avoid explicit unit expansion under suitable cost representations [1, 2].

3 Exact regularization thresholds

Assume that x^0 uniquely minimizes $c \cdot x$ over P . Total unimodularity makes x^0 an integer vertex, hence $x^0 \in P_q$ for every q . Let $\mathcal{G}$ be its finite set of feasible signed simple cycles. For $g \in \mathcal{G}$, define

$$A_g = c \cdot g > 0, \quad D_g = -2 \sum_e b_e x_e^0 g_e, \quad S_g = \sum_e b_e g_e^2 > 0. \quad (4)$$

Unique linear optimality gives $A_g > 0$ because $x^0 + g$ is feasible. All minima over empty sets below equal $+\infty$.

Theorem 2 (Continuous and grid thresholds) *Define*

$$\tau = \min_{g: D_g > 0} \frac{A_g}{D_g}, \quad \theta_q = \min_{g: D_g > S_g/q} \frac{A_g}{D_g - S_g/q}. \quad (5)$$

The plan x^0 minimizes F_t over P exactly for $0 \leq t \leq \tau$, and over P_q exactly for $0 \leq t \leq \theta_q$. It is the unique grid optimum for $t < \theta_q$; a distinct grid optimum appears at $t = \theta_q$ when this threshold is finite. The continuous optimum remains unique at a finite τ . If $\tau < \infty$, then $\theta_q > \tau$ for every q , allowing $\theta_q = \infty$. If $\tau = \infty$, every θ_q is infinite.

Proof. Every feasible direction at x^0 decomposes into a nonnegative combination of cycles in $\mathcal{G}$: orient positive differences forward and negative differences backward, then decompose the resulting circulation. The directional derivative along g is $A_g - tD_g$. Convexity therefore gives continuous optimality exactly when these quantities are nonnegative, which yields τ .

For a grid step, the exact expansion is

$$q[F_t(x^0 + g/q) - F_t(x^0)] = A_g - tD_g + \frac{t}{q}S_g. \quad (6)$$

Lemma 1 gives the grid threshold and uniqueness claims for $t > 0$. At $t = 0$, unique linear optimality applies. Strict convexity gives continuous uniqueness for $t > 0$. Finally, every eligible cycle satisfies

$$\frac{A_g}{D_g - S_g/q} > \frac{A_g}{D_g} \geq \tau.$$

There are finitely many cycles, so their minimum is strictly greater than τ whenever finite. If no D_g is positive, neither threshold can be finite. $\square$

The correction tS_g/q in (6) is the cost of a finite step beyond its directional derivative. It delays the first grid transition and can change which cycle determines that transition. For a fixed t , negative-cycle detection checks the threshold inequalities without listing all simple cycles. The result concerns persistence of x^0 ; it does not assert monotonicity of each flow along the whole path.

Example 1 (A sharp two-by-two instance) Let $\mu = \nu = (1, 1)$, $b_{ij} = 1$, and $c = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Write $x(s) = \begin{pmatrix} 1-s & s \\ s & 1-s \end{pmatrix}$. The unique linear optimum is $x(0)$ and

$$F_t(x(s)) = 2s + t(2 - 4s + 4s^2).$$

The only feasible cycle at $x(0)$ has $(A, D, S) = (2, 4, 4)$. Hence

$$\tau = \frac{1}{2}, \quad \theta_1 = \infty, \quad \theta_q = \frac{q}{2(q-1)} \quad (q \geq 2). \quad (7)$$

For $t > 0$, the continuous optimum has $s^{\mathbb{R}}(t) = \max\{0, \frac{1}{2} - \frac{1}{4t}\}$. Grid optima are the nearest feasible grid points to this value, with both choices retained at ties. Figure 1 shows the paths.

3.1 Pure group assignments

Fix $t > 0$ and write $a_e = tb_e$, so $F_t(x) = \sum_e (c_e x_e + a_e x_e^2)$. Suppose $n = m = M$ and ζ is a permutation with $\nu_{\zeta_i} = \mu_i$. Define

$$\bar{x}_{i\zeta_i} = \mu_i, \quad \bar{x}_{ij} = 0 \quad (j \neq \zeta_i).$$

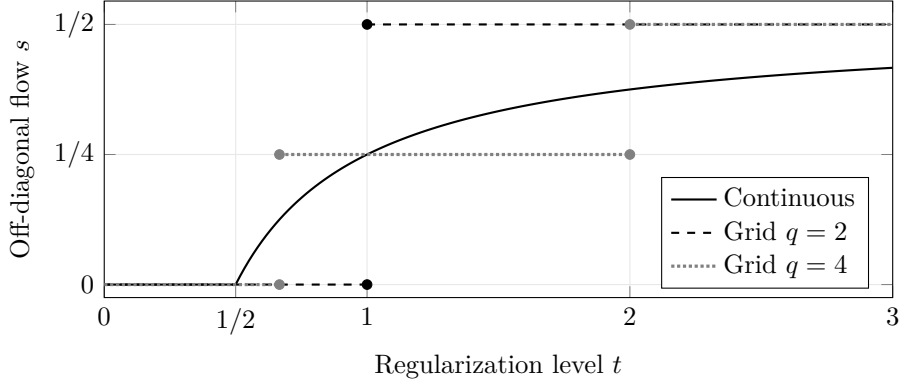


Fig. 1 Off-diagonal flow in Example 1. The continuous path leaves zero at $t = 1/2$, while the grids $q = 4$ and $q = 2$ leave at $2/3$ and 1 . Filled endpoints denote ties

The next characterization allows unequal group sizes and does not assume that $\bar{x}$ is linear optimal. For distinct groups i, k , set

$$v_{ik} = c_{i\zeta_k} - c_{i\zeta_i} - 2a_{i\zeta_i}\mu_i, \quad w_{ik}^{(q)} = v_{ik} + \frac{a_{i\zeta_k} + a_{i\zeta_i}}{q}. \quad (8)$$

Theorem 3 (Pure assignments) *The plan $\bar{x}$ is optimal over P_q if and only if every directed simple cycle of groups has nonnegative total $w^{(q)}$ -weight. It is uniquely optimal if and only if every such total is positive. In particular, the inequalities*

$$c_{ij} + \frac{a_{ij}}{q} \geq c_{i\zeta_i} + a_{i\zeta_i} \left(2\mu_i - \frac{1}{q} \right), \quad j \neq \zeta_i, \quad (9)$$

are sufficient, with strict inequalities implying uniqueness. Over P , the plan $\bar{x}$ is optimal if and only if every directed simple cycle has nonnegative total v -weight; whenever optimal, it is unique.

Proof. A group cycle moves $1/q$ units from each group's designated destination to the next group's destination. Its cost is q^{-1} times its total $w^{(q)}$ -weight. These are exactly the feasible nontrivial residual cycles at $\bar{x}$, so Lemma 1 proves the grid claims. For the continuous problem, their directional derivatives are the corresponding v -totals. Feasible directions decompose into these cycles, and convexity gives the criterion. Strict convexity gives uniqueness. $\square$

Corollary 4 (A sufficient linear-cost gap) *For each i , suppose $a_{ij} \geq a_{i\zeta_i}$ for $j \neq \zeta_i$ and*

$$\min_{j \neq \zeta_i} c_{ij} - c_{i\zeta_i} > a_{i\zeta_i} \mu_i^2 (1 - 1/M). \quad (10)$$

Then $\bar{x}$ is the unique optimum over P_1 .

Proof. Since $M \geq 2$, $\mu_i^2(1 - 1/M) \geq 2(\mu_i - 1)$, as $\mu_i^2/2 - 2(\mu_i - 1) = (\mu_i - 2)^2/2 \geq 0$. The assumptions therefore imply all strict inequalities in (9) with $q = 1$. $\square$

The coefficient ordering in this sufficient gap condition cannot simply be dropped. With $\mu = \nu = (2, 2)$ and

$$a = \begin{pmatrix} 10 & 1 \\ 1 & 10 \end{pmatrix}, \quad c = \begin{pmatrix} 1 & 22 \\ 22 & 1 \end{pmatrix},$$

the gap condition holds because $21 > 10 \cdot 4(1 - 1/2) = 20$, but the diagonal plan costs 84 and the all-ones plan costs 68. The exact criterion in Theorem 3 needs no coefficient ordering: here the two-cycle has total $w^{(1)}$ -weight -16 .

4 Replication and the threshold correction

The normalized grid formulation has a direct population interpretation. Multiplying the objective by q and putting $z = qx$ gives

$$\min_{x \in P_q} F_t(x) = \frac{1}{q} \min_{z \in qP \cap \mathbb{Z}^{n \times m}} \left\{ c \cdot z + \frac{t}{q} Q(z) \right\}. \quad (11)$$

Thus q replicates both margins, with congestion scaled by $1/q$ to keep the objective per replicated population comparable. With an unscaled coefficient on $Q(z)$, the continuous and integer thresholds are τ/q and θ_q/q .

Theorem 5 (Exact replication law) *Suppose $\tau < \infty$. For g with $D_g > 0$, set*

$$\alpha_g = \frac{A_g}{D_g}, \quad \rho_g = \frac{S_g}{D_g}, \quad \rho_* = \min_{g: \alpha_g = \tau} \rho_g. \quad (12)$$

Then θ_q is nonincreasing in q , strictly decreasing once finite, and converges to τ . It is finite exactly when $q > \min_{g: D_g > 0} \rho_g$. For every integer $q > \rho_$,*

$$0 < \theta_q - \tau \leq \frac{\tau \rho_*}{q - \rho_*}. \quad (13)$$

Moreover, for all sufficiently large integers q ,

$$\theta_q = \frac{\tau}{1 - \rho_*/q}, \quad \lim_{q \rightarrow \infty} q(\theta_q - \tau) = \tau \rho_*. \quad (14)$$

Consequently the unscaled threshold difference satisfies $q^2(\theta_q/q - \tau/q) \rightarrow \tau \rho_$.*

Proof. Theorem 2 gives

$$\theta_q = \min_{g: q > \rho_g} \frac{\alpha_g}{1 - \rho_g/q}.$$

Each eligible expression is strictly decreasing in q , and eligibility expands with q . This proves monotonicity and the finiteness criterion. A cycle with $(\alpha_g, \rho_g) = (\tau, \rho_*)$ gives (13), hence convergence.

Among cycles with $\alpha_g = \tau$, the smallest ρ_g gives the smallest threshold whenever it is eligible. Every other cycle has a strictly larger limiting value $\alpha_g > \tau$. Since there are finitely many cycles, none of those can minimize for sufficiently large q . This proves the exact identity in (14); the limits follow by expansion. $\square$

The reciprocal form makes the changes of minimizing cycle explicit:

$$\frac{1}{\theta_q} = \max \left\{ 0, \max_{g \in \mathcal{G}} \left(\frac{D_g}{A_g} - \frac{S_g}{q A_g} \right) \right\}, \quad \frac{1}{\infty} = 0. \quad (15)$$

It is the restriction of a convex piecewise-affine function of the grid size $1/q$. The monotonicity in Theorem 5 therefore holds even though successive grid sets need not be nested. In Example 1, $\rho_* = 1$ and $q(\theta_q - \tau) \rightarrow 1/2$, so the first-order rate cannot be improved.

Example 2 (The critical cycle can change) Take $\mu = (2, 2)$, $\nu = (1, 1, 2)$, and

$$b = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \end{pmatrix}, \quad c = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 1 & 0 \end{pmatrix}, \quad x^0 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

The unique linear optimum x^0 has two feasible simple cycles. Their (A, D, S) values are $(2, 10, 6)$ and $(1, 6, 6)$, respectively. Thus $\tau = 1/6$ and $\rho_* = 1$, while

$$\theta_q = \begin{cases} 2q/(10q - 6), & q = 1, 2, \\ q/[6(q - 1)], & q \geq 3. \end{cases} \quad (16)$$

Both cycles attain the threshold at $q = 3$. The cycle determining the continuous threshold therefore need not determine the integer threshold on a coarse grid.

Remark 1 (Bounded network flows) The cycle argument also applies to a nonempty bounded flow polytope $\{x : Ax = d, \ell \leq x \leq u\}$ with network incidence matrix A and integer data. Use feasible signed residual cycles and a unique linear optimum. Replication must scale balances and both bounds; keeping capacities fixed would invalidate (11).

5 Approximation along the regularization path

Write $V_q(t) = \min_{P_q} F_t$ and $V_{\mathbb{R}}(t) = \min_P F_t$. The next bound is a quadratic consequence of feasible dependent rounding [6]. It complements the first-transition formulas by controlling the entire path.

Proposition 6 (Value and plan bounds) *Fix $t > 0$, let $x^{\mathbb{R}} = x^{\mathbb{R}}(t)$, and put $r_e = qx_e^{\mathbb{R}} - \lfloor qx_e^{\mathbb{R}} \rfloor$. Then*

$$0 \leq V_q(t) - V_{\mathbb{R}}(t) \leq \frac{t}{q^2} \sum_e b_e r_e (1 - r_e) \leq \frac{t}{4q^2} \sum_e b_e. \quad (17)$$

For every grid optimum x^q ,

$$\sum_e b_e (x_e^q - x_e^{\mathbb{R}})^2 \leq \frac{V_q(t) - V_{\mathbb{R}}(t)}{t} \leq \frac{1}{4q^2} \sum_e b_e. \quad (18)$$

The value bound is attained or improved by a feasible rounding satisfying $\lfloor qx^{\mathbb{R}} \rfloor \leq qx \leq \lceil qx^{\mathbb{R}} \rceil$. Its least-cost choice is a binary transportation problem.

Proof. The polytope

$$\{z \in qP : \lfloor qx^{\mathbb{R}} \rfloor \leq z \leq \lceil qx^{\mathbb{R}} \rceil\}$$

contains $qx^{\mathbb{R}}$ and has integer vertices by total unimodularity. Express $qx^{\mathbb{R}}$ as a convex combination of these vertices, and let Z have the resulting distribution. Then $Y = Z/q$ satisfies $\mathbb{E}Y = x^{\mathbb{R}}$ and $\text{Var}(Y_e) = r_e(1 - r_e)/q^2$. Therefore

$$\mathbb{E}F_t(Y) - F_t(x^{\mathbb{R}}) = \frac{t}{q^2} \sum_e b_e r_e (1 - r_e).$$

Some vertex costs no more than this average. Writing $Z = \lfloor qx^{\mathbb{R}} \rfloor + w$ leaves binary choices and integer residual margins; the cost increments are linear in w . Finally, continuous first-order optimality and quadratic expansion give

$$F_t(x^q) - F_t(x^{\mathbb{R}}) = \nabla F_t(x^{\mathbb{R}}) \cdot (x^q - x^{\mathbb{R}}) + t \sum_e b_e (x_e^q - x_e^{\mathbb{R}})^2 \geq t \sum_e b_e (x_e^q - x_e^{\mathbb{R}})^2.$$

This proves (18). $\square$

The distance bound is uniform over $t > 0$ and over all grid optima. It does not imply that the best rounding is a global grid optimum. In Example 1, take $t = 1$. Then $x^{\mathbb{R}} = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}$. For every $q \equiv 2 \pmod{4}$, nearest grid points have $|s - s^{\mathbb{R}}| = 1/(2q)$, and

$$V_q(1) - V_{\mathbb{R}}(1) = \frac{1}{q^2}, \quad \sum_e b_e (x_e^q - x_e^{\mathbb{R}})^2 = \frac{1}{q^2}. \quad (19)$$

Since $\sum_e b_e = 4$, both constants in (17)–(18) are sharp. The same instance has a positive threshold discrepancy of order $1/q$. Thus a second-order value approximation can coexist with a first-order error in the regularization level at which the allocation changes.

There is also a simple condition for exact integrality. If the positive support of $x^{\mathbb{R}}(t)$ is a forest, a leaf's integer margin determines its only incident flow as an integer; successive leaf removal proves $x^{\mathbb{R}}(t) \in P_1$. For fixed $t > 0$ and b , a continuous integer optimum z with a positive-support cycle g instead requires

$$\sum_e g_e (c_e + 2tb_e z_e) = 0.$$

These equalities form a finite union of proper hyperplanes in c . Hence continuous optima that are integral and have cyclic positive support are exceptional with respect to linear costs.

6 Coordinate response thresholds

For completeness, consider the integer grid P_1 and fixed positive coefficients a_e . Vary only the linear cost of connection e , writing $c_e = s$. Define

$$H_e(k) = \min_{x \in P_1 : x_e = k} \left\{ a_e x_e^2 + \sum_{f \neq e} (c_f x_f + a_f x_f^2) \right\}. \quad (20)$$

For $e = (i, j)$, the feasible counts run from $L_e = \max\{0, \mu_i + \nu_j - \sum_h \mu_h\}$ to $U_e = \min\{\mu_i, \nu_j\}$. These bounds are attainable, and every intermediate integer count extends to a feasible table by completing its row and column and then the remaining margins.

Proposition 7 (Unit adjustments) *For $L_e < k < U_e$,*

$$H_e(k-1) + H_e(k+1) - 2H_e(k) \geq 2a_e. \quad (21)$$

Set $\sigma_k = H_e(k-1) - H_e(k)$ for $L_e < k \leq U_e$, with $\sigma_{L_e} = +\infty$ and $\sigma_{U_e+1} = -\infty$. The optimal count is k exactly when $\sigma_{k+1} \leq s \leq \sigma_k$. At a finite threshold σ_k , precisely $k-1$ and k are optimal counts, and consecutive finite thresholds are separated by at least $2a_e$. Thus, for $s_1 < s_2$ and any associated optimal counts k_1, k_2 ,

$$0 \leq k_1 - k_2 \leq 1 + \frac{s_2 - s_1}{2a_e}. \quad (22)$$

Proof. Let x, y attain $H_e(k-1), H_e(k+1)$ and denote the objective in (20) by Φ . A cycle decomposition of $y - x$ contains a signed simple cycle g with $g_e = 1$ conforming to its signs. Both $x + g$ and $y - g$ are feasible and have coordinate e equal to k . Quadratic expansion gives

$$\Phi(x) + \Phi(y) - \Phi(x + g) - \Phi(y - g) = 2 \sum_{f: g_f \neq 0} a_f (|y_f - x_f| - 1) \geq 2a_e.$$

Since $2H_e(k) \leq \Phi(x + g) + \Phi(y - g)$, this proves (21). The differences of $H_e(k) + sk$ are strictly increasing, so the stated neighboring inequalities characterize its minimum. A decline by $d \geq 1$ crosses d thresholds separated by at least $2a_e$; hence $s_2 - s_1 \geq 2a_e(d - 1)$. $\square$

This is a quantitative exchange argument for a convex flow value function, consistent with the established discrete convexity theory [3, 4]. Basic monotonicity needs less structure: adding the two optimality inequalities for $G(x) + sT(x)$ gives $(s_2 - s_1)[T(x^2) - T(x^1)] \leq 0$. Taking $T(x) = x_e$ or x_e^2 shows that increasing a connection's linear or quadratic coefficient weakly reduces its flow. Taking $T = Q$ shows that total quadratic congestion weakly decreases with t , although individual flows need not be monotone.

A Additional structural results

A.1 Existence and the number of feasible plans

Proposition A.1 (Existence and finite grids) *For balanced positive integer margins with total N , the polytope P is nonempty and compact, and each P_q is nonempty and finite. In particular,*

$$|P_q| \leq \prod_{i=1}^n \binom{q\mu_i + m - 1}{m - 1} \leq m^{qN}. \quad (23)$$

Every continuous objective attains its minimum on P , and every real-valued objective attains its minimum on P_q . Strict convexity gives uniqueness on P . A linear objective has an integral optimum.

Proof. The matrix $\mu\nu^T/N$ proves continuous feasibility; successively matching available integer supply and demand constructs a plan in P_1 . The constraints are closed and $0 \leq x_{ij} \leq \min\{\mu_i, \nu_j\}$, proving compactness. Counting nonnegative compositions of each row of qx while ignoring column restrictions gives the product in (23). For an integer $r \geq 1$, $\binom{r+m-1}{m-1} = \prod_{h=1}^r (h+m-1)/h \leq m^r$. The remaining claims follow from compactness, finiteness, strict convexity, and total unimodularity, respectively. $\square$

Continuous feasibility and compactness also hold for balanced positive real margins. They give existence for linear, entropic, and quadratic objectives, with $0 \log 0 = 0$ in the entropic case. Entropic regularization with positive coefficient and quadratic regularization with positive weights are strictly convex. A linear problem can have fractional optimal plans alongside integral ones, and a strictly convex objective can have several grid optima.

A.2 Continuous solution and smooth sensitivity

Consider $F(x) = c \cdot x + \sum_e a_e x_e^2$, $a_e > 0$; for the main model, $a_e = tb_e$ at fixed $t > 0$. Vectorize x , set $H = 2 \operatorname{diag}(a)$, and let $\tilde{R} \in \mathbb{R}^{(n+m) \times nm}$ contain all row and column equations. Let R omit the last column equation and put $d = (\mu_1, \dots, \mu_n, \nu_1, \dots, \nu_{m-1})^T$.

Proposition A.2 (Redundancy and the reduced system) *Writing $v = (\mathbf{1}_n, -\mathbf{1}_m)^T$, the full systems satisfy*

$$\ker(\tilde{R}H^{-1}\tilde{R}^T) = \text{span}\{v\}, \quad \text{rank}(\tilde{R}H^{-1}\tilde{R}^T) = n + m - 1, \quad (24)$$

$$\ker \begin{pmatrix} H & -\tilde{R}^T \\ -\tilde{R} & 0 \end{pmatrix} = \text{span}\{(0, v)\}. \quad (25)$$

The matrix $S = RH^{-1}R^T$ is positive definite, and the reduced KKT matrix is nonsingular. The equality-constrained minimizer is

$$\hat{x} = H^{-1} \left[R^T S^{-1} (d + RH^{-1}c) - c \right]. \quad (26)$$

If $\hat{x} \geq 0$, it is the unique minimizer over P .

Proof. The equations $\tilde{R}^T(u, w) = 0$ are $u_i + w_j = 0$ for every connection, so $\ker \tilde{R}^T = \text{span}\{v\}$. Positive definiteness of H gives (24). If (z, p) belongs to the full KKT nullspace, then $H z = \tilde{R}^T p$ and $\tilde{R} z = 0$; multiplying by z^T gives $z = 0$, then $p \in \text{span}\{v\}$. Deleting one redundant equation leaves R with full row rank, so $S \succ 0$ and the reduced KKT matrix is nonsingular. Solving $Hx + c = R^T p$, $Rx = d$ yields (26). If this vector is nonnegative, it also minimizes over P . $\square$

Thus singularity of the full system reflects only the multiplier normalization $(u, w) \mapsto (u + \alpha \mathbf{1}_n, w - \alpha \mathbf{1}_m)$. It does not obstruct an explicit solution or local comparative statics. If $\hat{x}$ has negative entries, the active nonnegativity constraints must be included; coordinatewise clipping need not preserve the margins.

Corollary A.3 (Local derivatives and the envelope) *On a region where the continuous optimum is strictly positive, define*

$$K = H^{-1} - H^{-1}R^T S^{-1}RH^{-1} \succeq 0. \quad (27)$$

With $\mathbf{e}_e$ the coordinate vector for connection e ,

$$\frac{\partial x}{\partial c} = -K, \quad \frac{\partial x}{\partial a_e} = -2x_e K \mathbf{e}_e, \quad \frac{\partial x}{\partial d} = H^{-1}R^T S^{-1}. \quad (28)$$

In particular, $\partial x_e / \partial c_e \leq 0$ and $\partial x_e / \partial a_e \leq 0$. For $V(a, c) = \min_{x \in P} F(x)$, including at boundary optima,

$$\frac{\partial V}{\partial c_e} = x_e, \quad \frac{\partial V}{\partial a_e} = x_e^2. \quad (29)$$

Proof. Differentiating the reduced KKT equations gives (28). Also $K = H^{-1/2}(I - \mathcal{P})H^{-1/2}$, where $\mathcal{P}$ is the orthogonal projector onto the row space of $RH^{-1/2}$. Thus $K_{ee} \geq 0$, giving the own-cost signs. The envelope identities follow from unique minimization on the fixed compact set P ; they do not require an interior optimum. $\square$

These are the reduced equality-constrained quadratic formulas [8, Chapters 5 and 10]. On a fixed active face, the same calculation uses its free variables and independent constraints. Margin perturbations must preserve feasibility on that face. Changes of support can introduce kinks in the optimizer. For the main parametrization, the envelope also gives $\partial V_{\mathbb{R}} / \partial b_e = t(x_e^{\mathbb{R}})^2$ and $dV_{\mathbb{R}}/dt = Q(x^{\mathbb{R}})$ for $t > 0$.

A.3 The general two-by-two solution

Proposition A.4 (Closed-form continuous and grid solutions) *For $n = m = 2$, every feasible plan has the form*

$$x(s) = \begin{pmatrix} s & \mu_1 - s \\ \nu_1 - s & \mu_2 - \nu_1 + s \end{pmatrix}, \quad L \leq s \leq U,$$

where $L = \max\{0, \nu_1 - \mu_2\}$ and $U = \min\{\mu_1, \nu_1\}$. Put

$$\begin{aligned}\kappa &= a_{11} + a_{12} + a_{21} + a_{22}, \\ \ell &= c_{11} - c_{12} - c_{21} + c_{22} - 2a_{12}\mu_1 - 2a_{21}\nu_1 + 2a_{22}(\mu_2 - \nu_1).\end{aligned}\tag{30}$$

Then the continuous minimizer is $x(s^{\mathbb{R}})$, where

$$s^{\mathbb{R}} = \min\{U, \max\{L, -\ell/(2\kappa)\}\}.\tag{31}$$

The optimizers over P_q are the plans whose s is nearest to $s^{\mathbb{R}}$ in $q^{-1}\mathbb{Z} \cap [L, U]$, with both choices included at a tie.

Proof. Substitution gives $F(x(s)) = \kappa s^2 + \ell s + \text{constant}$. Since $\kappa > 0$, its vertex is $-\ell/(2\kappa)$; clipping gives the continuous minimizer. The endpoints L, U belong to every grid because the margins are integer. Completing the square therefore gives the stated grid rule, including when the vertex lies outside $[L, U]$. $\square$

The formula covers both interior and boundary allocations. With $a_{ij} = 2$, $\mu = (20, 20)$, $\nu = (12, 28)$, and $c = \begin{pmatrix} 24 & 48 \\ 16 & 24 \end{pmatrix}$, it gives $\kappa = 8$, $\ell = -112$, and the common continuous and integer optimum $\begin{pmatrix} 7 & 13 \\ 5 & 15 \end{pmatrix}$. With $a = c = \begin{pmatrix} 200 & 2 \\ 2 & 200 \end{pmatrix}$ and $\mu = \nu = (10, 10)$, the unconstrained vertex is $-79/202$, so the common optimum is $\begin{pmatrix} 0 & 10 \\ 10 & 0 \end{pmatrix}$. Online Resource 1 also verifies a four-by-four pure assignment using Theorem 3.

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Statements and Declarations

All examples use synthetic data specified in the article. Exact verification scripts are provided in Online Resource 1; no external dataset is required.

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